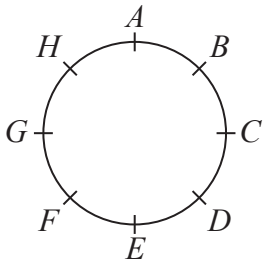
18 Hasmeet walks once round a circle with diameter 80 metres.



There are 8 points equally spaced on the circumference of the circle.

(a) Find the distance Hasmeet walks between one point and the next point.

Circumference = π × diameter

= π × 80 = 80π metres

1 mark
Use the circumference formula

Eight equal arcs: distance = 80π ÷ 8 = 10π metres.

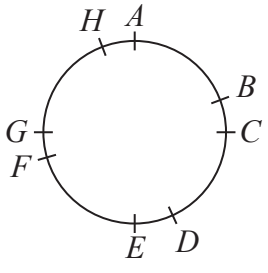
10π = 31.4159... m, so about 31.4 m.

1 mark
Correct arc length; accepted 31.4–31.5 m or 10π

31.4 m

(2)

Four of the points are moved, as shown in the diagram below.



Hasmeet walks once round the circle again.

(b) Has the mean distance that Hasmeet walks between one point and the next point changed? You must give a reason for your answer.

No. The total distance is still 80π and there are still 8 gaps, so the mean is still 80π ÷ 8 = 10π metres.

1 mark
Same total distance and number of gaps

(1)

(Total for Question 18 is 3 marks)